

COMPSCI 1DM3 - Winter 2026

Discrete Mathematics for Computer Science

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Last Lecture:

- Given two sets A and B , we defined:
 - Union: $A \cup B = \{x \mid x \in A \vee x \in B\}$
 - Intersection: $A \cap B = \{x \mid x \in A \wedge x \in B\}$
 - Difference: $A - B = \{x \mid x \in A \wedge x \notin B\}$
 - Complement: $\bar{A} = U - A = \{x \mid x \notin A\}$
 - Symmetric difference: $A \oplus B = (A - B) \cup (B - A)$
- We showed that $A \oplus B = (A \cup B) - (A \cap B)$
- We showed that $A - B = A \cap \bar{B}$

Last Lecture:

- De Morgan's laws:

$$\overline{(A \cap B)} = \overline{A} \cup \overline{B}$$

$$\overline{(A \cup B)} = \overline{A} \cap \overline{B}$$

- Distributive laws:

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

- $A \cup \overline{A} = U$, and $A \cap \overline{A} = \emptyset$

- Absorption laws:

$$A \cup (A \cap B) = A$$

$$A \cap (A \cup B) = A$$

Section 2.3

Functions

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Definition.

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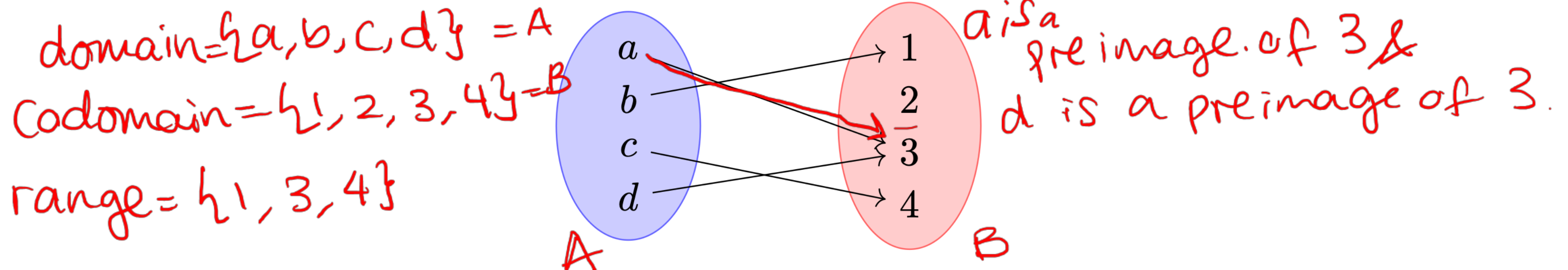
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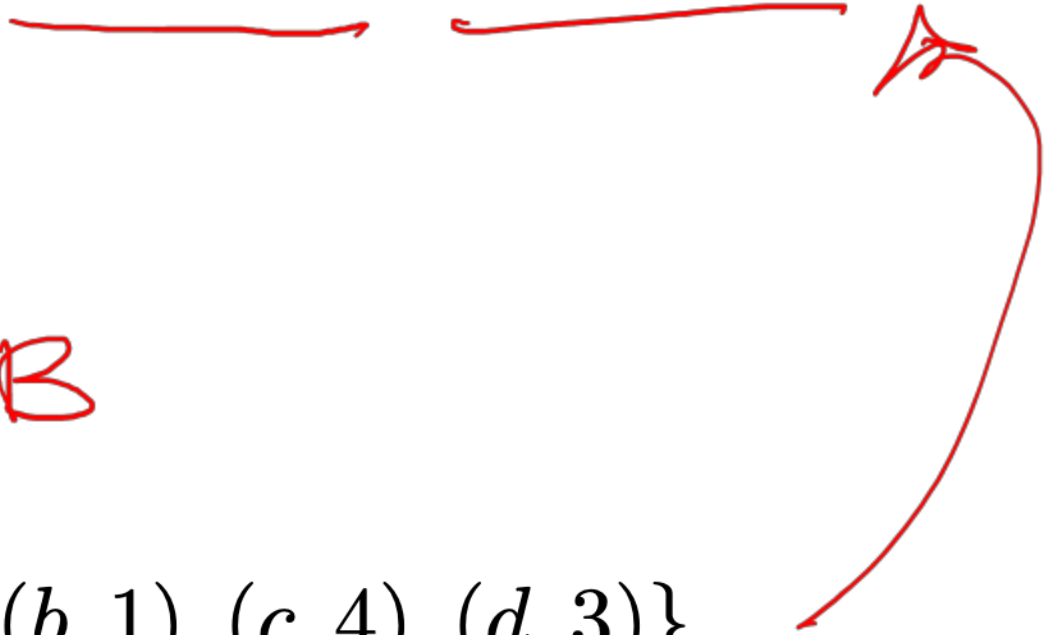
How to represent functions:

- Draw sets A and B and all the mapping:



- Implicit representation:

Example:

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad y = f(x), \quad f(x) = x^2$$
$$y - x^2 = 0$$


- Set representation:

Example:


$$\{(a, 3), (b, 1), (c, 4), (d, 3)\}$$
$$f(a) = 3, f(b) = 1, \dots$$


Some terminology

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- Range:

$$f : A \rightarrow B,$$

The range of f is $\{f(x) \mid x \in A\}$.

Some terminology

- Image of a set:

$$f : A \rightarrow B,$$

The image of a set $X \subseteq A$, denoted by $\underline{f(X)}$, is defined as $\{f(x) \mid x \in X\}$.

Example

Let $f : \mathbb{Z} \rightarrow \mathbb{R}$ be given by $f(x) = x^2$. What are the domain and codomain? Range? $f(\{1, 2, 3, 4\})$? What is the image of 3? preimage of 144?

domain: $\mathbb{Z}$

codomain: $\mathbb{R}$

range: $\{x^2 \mid x \in \mathbb{Z}\}$
 $= \{0, 1, 4, \dots\}$

$$f(\{1, 2, 3, 4\}) = \{1, 4, 9, 16\}$$

Image of 3: 9

Preimage of 144: $\{-12, 12\}$

Definition

- We say function f and g are equal, denoted by $f = g$, if
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Example. Consider real-valued functions $f(x) = 1 - x^2$ and $g(x) = x^2$. What are $f + g$ and fg ?

$$f + g = f(x) + g(x) = 1 - x^2 + x^2 = 1$$

$$(fg) = f(x)g(x) = (1 - x^2)x^2$$

Injective

A function $f : A \rightarrow B$ is called *one-to-one* (or *injective*) if for all $a, b \in A$

$$\text{if } f(a) = f(b) \text{ then } a = b$$

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or equivalently

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Example.

Is $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = x^2$ injective? $\rightarrow$ No

$$x_1, x_2 \in \mathbb{R}$$

$$f(x_1) = f(x_2)$$

$$\sqrt{x_1^2} = \sqrt{x_2^2}$$

$$\pm x_1 = \pm x_2$$

Counterexample: $x_1 = 1, x_2 = -1$

$$f(x_1) = 1 = f(x_2)$$

$$x_1 \neq x_2$$

Sufficient condition for injectivity

- A function f is called *increasing* if

$$\forall x, y \in \mathbb{R} \quad \text{if } x < y \quad \text{then } \underline{f(x) \leq f(y)}$$

- ~~•~~ A function f is called *strictly increasing* if

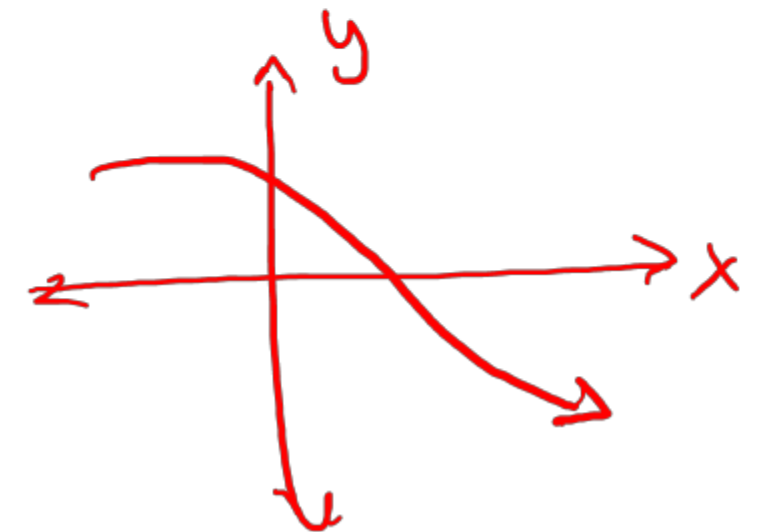
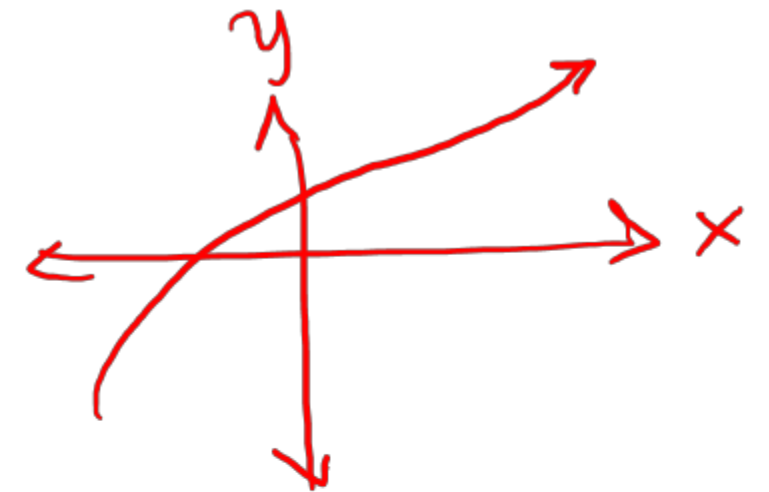
$$\forall x, y \in \mathbb{R} \quad \text{if } x < y \quad \text{then } \underline{f(x) < f(y)}$$

- A function f is called *decreasing* if

$$\forall x, y \in \mathbb{R} \quad \text{if } x < y \quad \text{then } f(x) \geq f(y)$$

- A function f is called *strictly decreasing* if

$$\forall x, y \in \mathbb{R} \quad \text{if } x < y \quad \text{then } f(x) > f(y)$$



Sufficient condition for injectivity

Theorem

Every strictly increasing function f is injective.

x_1, x_2 in domain, if $f(x_1) = f(x_2) \rightarrow x_1 = x_2$

Contradiction:

Let's assume $x_1 \neq x_2$

$x_1 < x_2$

f is strictly inc.

$f(x_1) < f(x_2)$

↳ contradiction!

$x_2 < x_1$

$f(x_2) < f(x_1)$

↳ contradicting

x_1 must be equal to $x_2 \rightarrow f$ is injective $\square$

Examples

Determine whether the following functions are injective:

- $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = 2x + 1$

$f(x)$ is strictly increasing.

$$x, y \in \mathbb{R}, \text{ if } \underline{x < y} \rightarrow \begin{aligned} &\cancel{2} f(x) < f(y) \\ &2x + 1 < 2y + 1 \end{aligned}$$

$\rightarrow f$ is injective

✓

Examples

Determine whether the following functions are injective:

- $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = 2x + 1$

- $f : \mathbb{R} \setminus \{-1/2\} \rightarrow \mathbb{R}$ given by $f(x) = \frac{1}{2x+1}$ $f(x_1) = f(x_2) \rightarrow x_1 = x_2$
 ~~$x, y \in \mathbb{R} \setminus \{-1/2\}$~~
 $f(x_1) = \frac{1}{2x_1+1} = \frac{1}{2x_2+1} = f(x_2)$
 $2x_2+1 = 2x_1+1$
 $2x_2 = 2x_1 \rightarrow x_2 = x_1$

Examples

Determine whether the following functions are injective:

- $f : \mathbb{R}^+ \rightarrow \mathbb{R}$ given by $f(x) = \log(x)$

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- $f : \mathbb{R}^+ \rightarrow \mathbb{R}$ given by $f(x) = \log(x)$

$$x_1, x_2 \in \mathbb{R}^+$$

$$\log(x_1) = \log(x_2)$$

$$, \text{ if } x_1 < x_2$$

$$\log(x_1) < \log(x_2)$$

$$\text{bc. str. inc.} \rightarrow \log(x) \text{ inc}$$

- $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = |x|$

$$x_1 = -2$$

$$|-2| = 2 = |2|$$

$$x_2 = 2$$

$$x_1 \neq x_2$$

Not.